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Space Mechanics – SP423

Design of a New Mission to Neptune

Theoretical study and numerical simulation with STK 13

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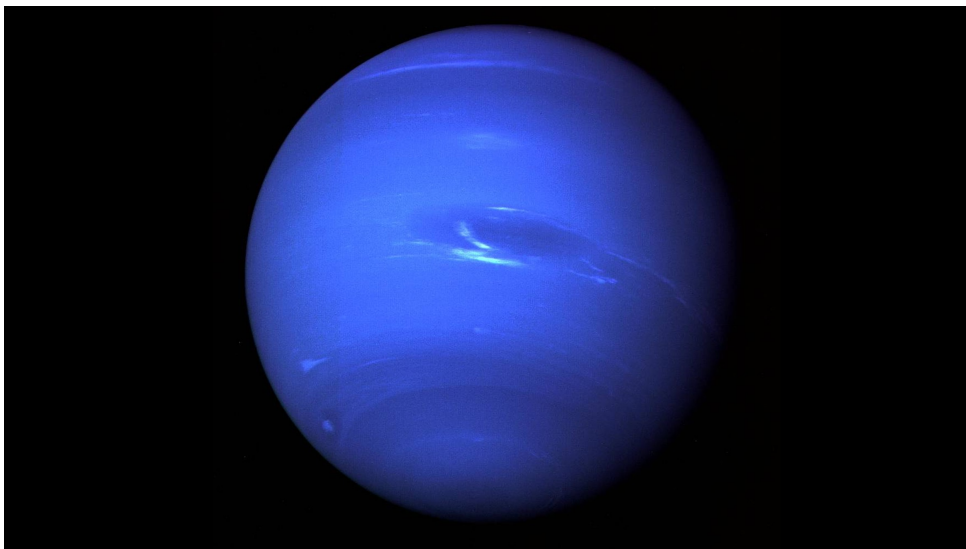
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Abstract

This report presents the preliminary design of an interplanetary mission from Earth to Neptune, carried out in the framework of the IPSA Satellite Company (ISC). The mission objective is twofold: estimate the cost (in terms of ΔV) of sending a new spacecraft to Neptune, and identify the most suitable launch period. The work follows a two-step methodology: first, an initial trajectory is selected using the NASA Trajectory Browser and confirmed analytically using the patched conics approximation. Second, two high-fidelity numerical simulations are carried out with STK and the Astrogator module to validate the theoretical results and explore two distinct design choices. A gravity assist at Jupiter is exploited to reduce the propellant budget. The mission departs from Earth on 23 February 2031, performs a Jupiter flyby on 21 July 2032 and reaches Neptune between October 2037 (Simulation 2, fast cruise with a powered Oberth-boosted swing-by) and May 2039 (Simulation 1, slow optimised cruise with a ballistic flyby). The total ΔV to *reach* Neptune ranges from ~ 7.29 km/s (NASA Trajectory Browser reference) to ~ 8.87 km/s (Simulation 2 with a powered swing-by), confirming the feasibility of the mission with a heavy launcher of the Ariane 6 or Falcon Heavy class.

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1. Introduction

In this project, we will work on the determination of the best trajectory to send a satellite into orbit around Neptune. To do this, we are part of the IPSA Satellite Company, and we have been tasked with designing a space mission.

Several similar missions were launched in the 1960s and 70s, focusing on the outer planets of the Solar System, but Neptune was never the primary focus; the studies were mainly concentrated on Mars, Jupiter, and Saturn. Twenty-seven years ago, Voyager 2 performed a flyby of Neptune and its moons, and it remains the only spacecraft to have visited the planet. Another space probe project to Neptune, *Neptune Orbiter*, was proposed to NASA in 2005; others were subsequently proposed, but none was ever approved. More recently, in 2021, ESA published a White Paper requesting renewed scientific interest in the ice giants Uranus and Neptune, which constitutes the premise of a future mission.

Therefore, the objective of this study is to calculate the cost of a mission conducted by a new spacecraft making a journey from Earth to Neptune and to define the most suitable period for such a mission. To answer these two questions, the work is divided in four parts, in agreement with the project specification:

1. Direct application of the lectures to determine the theoretical interplanetary transfer from Earth to Neptune, with gravitational assistance at our convenience;
2. Numerical simulation with STK software, using some theoretical characteristics as constraint parameters;
3. Comparison and critical analysis of both sets of results;
4. Answers to the two critical questions on cost and launch period.

2. Trajectory search

To find the best trajectory for this mission we first used the NASA Trajectory Browser tool. Using the *Trajectory search* option we input the destination (Neptune) and the proper trajectory constraints to match the mission parameters.

We chose a one-way flyby mission type because we do not plan for the satellite to have a return trajectory back to Earth after its orbit around Neptune. The trajectory search will find us a flyby trajectory that we will use as the basis for our trajectory, that will later include a capture burn around Neptune in order to stay in orbit around it.

We have a launch window starting in 2026 as imposed by the project (the launch date must be later than 14 November 2026) and we put the maximum value for the *Max duration* and *Delta-V* values, so as to obtain the largest number of options.

Trajectory Search

This page features a tool to find spacecraft trajectories to planets and small-bodies. For ideas of what to search for, view the [example queries page](#). A [user guide](#) is also available explaining search constraints, the methodology behind trajectory computations and the important limitations of the tool.

Select Destinations

☐ NEOs with constraints:

☐ Asteroids ☐ Comets

☐ Max absolute magnitude of

☐ Max orbit condition code of

☒ Custom list

Near-Earth Objects (NEOs) are bodies that orbit the sun with perihelion distance less than 1.3 AU. [\[more\]](#)

A body's absolute magnitude gives a rough idea of its size. For example, a magnitude 25 object is approximately 35 m in size whereas a magnitude 20 object is about 350 m in diameter. [\[more\]](#)

The orbit condition code is an integer between 0 and 9 indicating how well an object's orbit is known, where 0 indicates a well-determined orbit and 9 insufficient observations for long-term orbit prediction. [\[more\]](#)

Alternatively include a custom list of NEOs or planets, one object per line. E.g. 'Apophis', 'Mars', or '2010 UE51'. [\[more\]](#)

A list of NEOs matching specific parameters of interest can be generated with the [Small-Body database search engine](#).

Specify Trajectory Constraints

Mission type:

Launch year: -

Max duration: years

Max ΔV : km/s

Minimize:

Show:

Specify if the trajectory is to include an Earth-return segment and indicate nature of encounter. [\[more\]](#)

Range of acceptable launch years, assumed to refer to January 1st of bounding years. [\[more\]](#)

Maximum acceptable total mission duration. [\[more\]](#)

Maximum total ΔV (indicator of propellant requirement) from a 200 km circular Low-Earth Orbit (LEO). [\[more\]](#)

Parameter to minimize if more than one trajectory per destination is found to meet above constraints. [\[more\]](#)

Specify maximum frequency of optima per destination across launch year window. [\[more\]](#)

Figure 2.1: NASA Trajectory Browser search page and parameters used for our query.

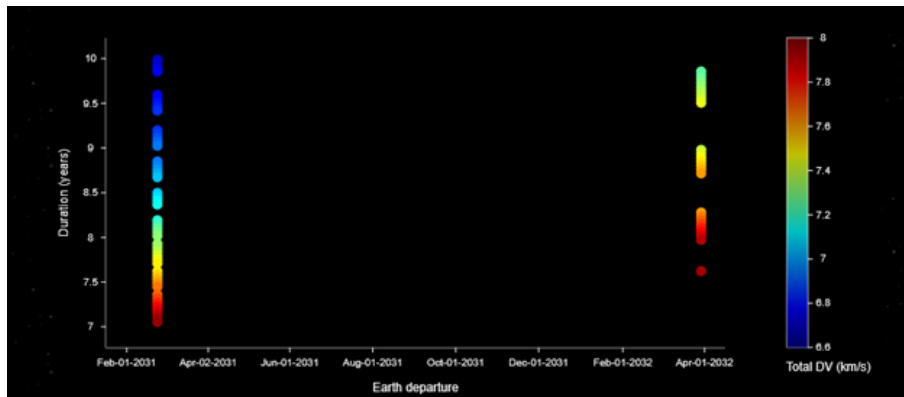


Figure 2.2: Graphic visualization of the results returned by the NASA Trajectory Browser.

We exported the trajectories as a CSV file, and we plotted the following graph to visualize and compare all the trajectories in order to find the one that fits best our needs.

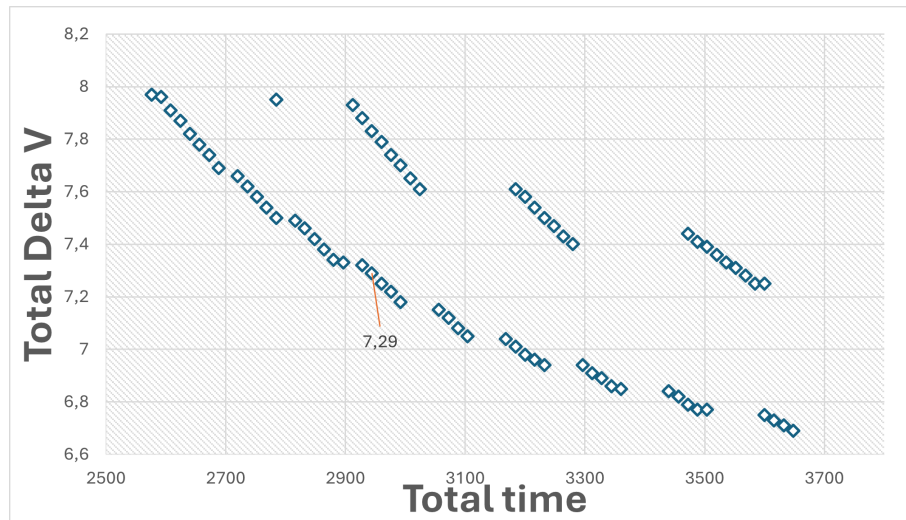


Figure 2.3: Total ΔV as a function of total mission time for all candidate trajectories.

We ended up choosing a trajectory that has a balance between the mission time and the total ΔV necessary. The trajectory we chose demands 7.29 km/s of total ΔV and will take 8.06 years. The mission plan is summarised below: the craft will leave Earth on 23 February 2031, where it will perform an impulsive burn of 7.17 km/s to escape Earth's sphere of influence (SOI) and enter a heliocentric orbit that will lead it to a flyby of Jupiter on 19 July 2032 for a gravity assist, followed by a 114 m/s correction burn for its final trajectory to Neptune, where it will arrive on 17 March 2039.

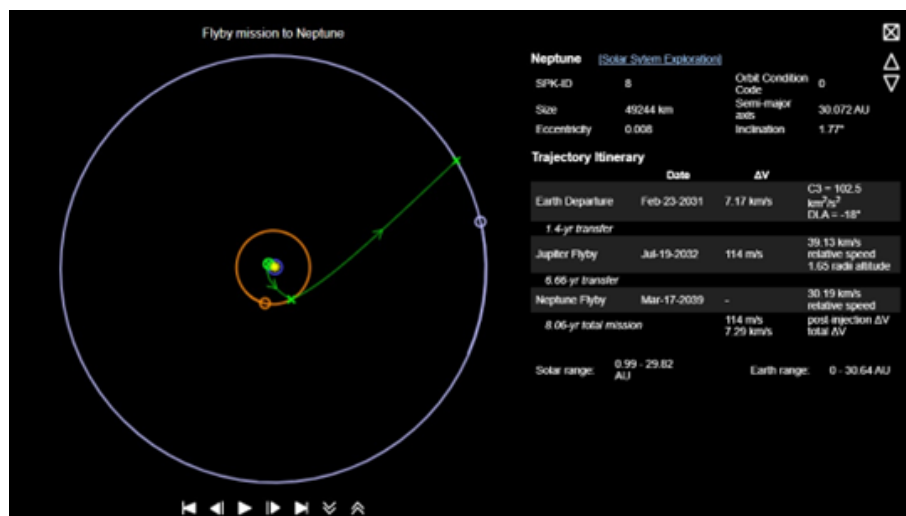


Figure 2.4: Visualisation of the selected flyby mission from Earth to Neptune via a Jupiter gravity assist.

Table 2.1: Summary of the selected baseline trajectory (NASA Trajectory Browser, flyby).

Parameter	Value
Launch date (Earth)	23 February 2031
Jupiter flyby date	19 July 2032
Neptune arrival date	17 March 2039
Earth departure burn ΔV_1	7.17 km/s
Mid-course correction ΔV_2	0.114 km/s
Total mission ΔV (flyby)	7.29 km/s
Total mission duration	8.06 years
Mission architecture	Flyby with Jupiter gravity assist

Remark. The NASA Trajectory Browser searches for *flyby* trajectories, i.e. it does not include the cost of a Neptune Orbit Insertion burn. In our STK simulation (Chapter 4), we extend this baseline by adding a capture manoeuvre at Neptune so as to enter a bound orbit around the planet. The corresponding ΔV is reported for reference but is not part of the mission budget, since the project asks only for the ΔV to *reach* Neptune.

3. Application of the lectures

3.1 Model hypotheses and notations

Hypotheses

In accordance with the hypotheses stated in the course (*Slide 2 – Framework*):

- (1) All planets (except Pluto) orbit in the ecliptic plane.
- (2) All planets are on circular orbits.
- (3) We use the *sphere of influence* (SoI) concept to break the trajectory into successive two-body problems.
- (4) The gravity assist is treated as a hyperbolic flyby in the planet frame (conservation of $\|\mathbf{V}_\infty\|$).

Main notations

μ_X	gravitational parameter of body X
r_X	Sun–planet X distance
R_X	physical radius of planet X
V_X	mean orbital velocity of X around the Sun
ρ_Σ	sphere of influence radius
$V_{\infty E}$	excess hyperbolic velocity at Earth SoI exit
ψ	angle between $\mathbf{V}_{\infty E}$ and $\mathbf{V}_E$
V_{0S}	heliocentric velocity at Earth SoI exit
γ_{0S}	flight path angle of $\mathbf{V}_{0S}$ relative to local horizontal
V_{S1}, γ_1	heliocentric velocity and flight path angle at Jupiter arrival
$V_{\infty 1}, V_{\infty 2}$	Jupiter-relative velocities before/after flyby
V_{S2}, γ_{S2}	heliocentric velocity and flight path angle after flyby
δ	flyby deflection angle
V_{S3}, γ_3	heliocentric velocity and flight path angle at Neptune arrival

3.2 Constants used

Quantity	Symbol	Value
Sun grav. parameter	μ_S	$1.32712 \times 10^{11} \text{ km}^3/\text{s}^2$
Earth grav. parameter	μ_E	$3.9860 \times 10^5 \text{ km}^3/\text{s}^2$
Jupiter grav. parameter	μ_J	$1.26687 \times 10^8 \text{ km}^3/\text{s}^2$
Neptune grav. parameter	μ_N	$6.8351 \times 10^6 \text{ km}^3/\text{s}^2$
Sun–Earth distance	r_E	$1.496 \times 10^8 \text{ km}$
Sun–Jupiter distance	r_J	$7.786 \times 10^8 \text{ km}$
Sun–Neptune distance	r_N	$4.495 \times 10^9 \text{ km}$
Earth radius	R_E	6 378 km
Jupiter radius	R_J	71 492 km
Earth orbital velocity	V_E	29.78 km/s (computed)
Jupiter orbital velocity	V_J	13.06 km/s (computed)
Neptune orbital velocity	V_N	5.43 km/s (computed)

The orbital velocities are obtained assuming circular orbits:

$$V_X = \sqrt{\frac{\mu_S}{r_X}}. \quad (3.1)$$

3.3 Phase 1 – Departure from Earth

Phase hypotheses: the probe is in a circular parking orbit at $z = 200$ km altitude, then a tangential impulse places it on a hyperbolic escape trajectory with a given asymptotic velocity $V_{\infty E}$.

3.3.1 Escape hyperbola

The periapsis radius of the hyperbola is

$$r_p = R_E + z = 6\,378 + 200 = 6\,578 \text{ km}. \quad (3.2)$$

The circular parking velocity at this altitude is

$$V_{\text{circ}} = \sqrt{\frac{\mu_E}{r_p}} = 7.784 \text{ km/s}. \quad (3.3)$$

From energy conservation along the hyperbolic trajectory,

$$\frac{V_{0E}^2}{2} - \frac{\mu_E}{r_p} = \frac{V_{\infty E}^2}{2} \iff V_{0E} = \sqrt{V_{\infty E}^2 + \frac{2\mu_E}{r_p}}. \quad (3.4)$$

With $V_{\infty E} = 11.053 \text{ km/s}$ ¹, we obtain:

$$V_{0E} = 15.600 \text{ km/s}, \quad \Delta V = V_{0E} - V_{\text{circ}} = 7.816 \text{ km/s}. \quad (3.5)$$

The escape hyperbola elements are:

$$a = \frac{\mu_E}{V_{\infty E}^2} = 3\,263 \text{ km} \quad (\text{magnitude}), \quad (3.6)$$

$$e = 1 + \frac{r_p V_{\infty E}^2}{\mu_E} = 3.016. \quad (3.7)$$

3.3.2 Hyperbolic velocity at SoI exit

According to course formula (5.26):

$$\boxed{V_{\infty E} = \sqrt{\frac{\mu_E}{a}}} \quad (3.8)$$

which corresponds to the asymptotic velocity on the hyperbolic trajectory. This is the velocity used to characterize the Earth SoI exit.

¹The value of $V_{\infty E}$ is tuned so that the total Jupiter–Neptune time equals 6.66 yr (see §3.7). NASA itself uses $V_{\infty E} \approx 10.12 \text{ km/s}$ ($C_3 = 102.5 \text{ km}^2/\text{s}^2$) by exploiting the 3D geometry of planetary orbits.

3.3.3 Time to exit the sphere of influence

The Earth SoI radius is defined by Laplace's formula (*course*, eq. 5.20):

$$\rho_\Sigma = \left(\frac{\mu_E}{\mu_S} \right)^{2/5} r_E \approx 805\,000 \text{ km}. \quad (3.9)$$

On a hyperbolic orbit, position as a function of the hyperbolic anomaly H is given by

$$r = a(e \cosh H - 1) \iff \cosh H = \frac{r/a + 1}{e}. \quad (3.10)$$

At SoI exit ($r = \rho_\Sigma$):

$$\cosh H = \frac{805\,000/3\,263 + 1}{3.016} = 82.14 \implies H = \operatorname{arccosh}(82.14) = 5.101 \text{ rad}. \quad (3.11)$$

Numerical trick. Excel does not have the ACOSH function, so we use the logarithmic identity:

$$\operatorname{arccosh}(x) = \ln(x + \sqrt{x^2 - 1}). \quad (3.12)$$

The hyperbolic Kepler equation reads (*course* eq. 5.25):

$$M = e \sinh H - H, \quad n = \sqrt{\frac{\mu_E}{a^3}}, \quad t = \frac{M}{n}. \quad (3.13)$$

Numerically:

$$\Delta M = 3.016 \times \sinh(5.101) - 5.101 = 242.61 \text{ rad}, \quad n = 3.388 \times 10^{-3} \text{ rad/s}, \quad (3.14)$$

$$t_{\text{SoI exit}} = \frac{242.61}{3.388 \times 10^{-3}} \approx 71\,612 \text{ s} \approx 19.9 \text{ h}. \quad (3.15)$$

3.4 Phase 2 – Heliocentric transfer Earth $\rightarrow$ Jupiter

At Earth SoI exit, the probe has Earth-relative velocity $\mathbf{V}_{\infty E}$, which adds vectorially to Earth's orbital velocity $\mathbf{V}_E$ to give the initial heliocentric transfer velocity $\mathbf{V}_{0S}$.

3.4.1 Heliocentric state at SoI exit

Velocity triangle (slide 10). With ψ the angle between $\mathbf{V}_{\infty E}$ and $\mathbf{V}_E$, the law of cosines gives:

$$V_{0S}^2 = V_{\infty E}^2 + V_E^2 + 2 V_{\infty E} V_E \cos \psi. \quad (3.16)$$

With $\psi = -18^\circ$ ($\mathbf{V}_{\infty E}$ points slightly “behind” $\mathbf{V}_E$):

$$V_{0S} = \sqrt{11.053^2 + 29.78^2 + 2 \times 11.053 \times 29.78 \cos(-18^\circ)} = 40.44 \text{ km/s}. \quad (3.17)$$

Flight path angle of $\mathbf{V}_{0S}$ (slide 11, law of sines). The flight path angle γ_{0S} (measured from the local horizontal, i.e., from $\mathbf{V}_E$) satisfies:

$$\frac{\sin \gamma_{0S}}{V_{\infty E}} = \frac{\sin \psi}{V_{0S}}. \quad (3.18)$$

Hence

$$\sin \gamma_{0S} = \frac{11.053 \times \sin(-18^\circ)}{40.44} = -0.0844 \implies \gamma_{0S} = -4.84^\circ. \quad (3.19)$$

The negative sign indicates that the probe has an inward radial component: it lies *before* the perihelion of the transfer orbit.

3.4.2 Characterization of the transfer orbit

Using the two invariants of Keplerian motion:

$$\varepsilon = \frac{V_{0S}^2}{2} - \frac{\mu_S}{r_E} \quad (\text{specific energy}) \quad (3.20)$$

$$h = V_{0S} r_E \cos \gamma_{0S} \quad (\text{specific angular momentum}) \quad (3.21)$$

Numerically,

$$\varepsilon = -69.37 \text{ km}^2/\text{s}^2 \quad (< 0, \text{ ellipse}), \quad (3.22)$$

$$h = 6.03 \times 10^9 \text{ km}^2/\text{s}. \quad (3.23)$$

We deduce the orbital elements of the transfer ellipse:

$$a = -\frac{\mu_S}{2\varepsilon} = 9.57 \times 10^8 \text{ km}, \quad e = \sqrt{1 + \frac{2\varepsilon h^2}{\mu_S^2}} = 0.845, \quad (3.24)$$

$$p = \frac{h^2}{\mu_S} = 2.74 \times 10^8 \text{ km}. \quad (3.25)$$

3.4.3 State at arrival at Jupiter

We apply energy conservation (*course eq. 5.29, slide 12*):

$$\boxed{\frac{V_{S1}^2}{2} - \frac{\mu_S}{r_J} = \frac{V_{0S}^2}{2} - \frac{\mu_S}{r_E}}. \quad (3.26)$$

Isolating V_{S1} :

$$V_{S1} = \sqrt{V_{0S}^2 + 2\mu_S \left(\frac{1}{r_J} - \frac{1}{r_E} \right)} = 14.22 \text{ km/s}. \quad (3.27)$$

And the conservation of angular momentum (*course eq. 5.30*):

$$\boxed{V_{S1} r_J \cos \gamma_1 = V_{0S} r_E \cos \gamma_{0S}}, \quad (3.28)$$

which gives the arrival flight path angle at Jupiter:

$$\cos \gamma_1 = \frac{h}{V_{S1} r_J} = 0.545 \quad \implies \quad \gamma_1 = 57.01^\circ. \quad (3.29)$$

3.4.4 Time of flight Earth $\rightarrow$ Jupiter

Since the orbit is elliptic ($e < 1$), we use the classical Kepler equation. We start with the true anomalies at Earth and Jupiter, obtained by inverting the polar equation $r = p/(1 + e \cos \nu)$:

$$\cos \nu = \frac{p/r - 1}{e}. \quad (3.30)$$

The *sign* of ν must match the sign of the radial velocity, i.e., the sign of $\sin \gamma$. At Earth, $\gamma_{0S} < 0$ so $\nu_0 < 0$; at Jupiter, $\gamma_1 > 0$ so $\nu_1 > 0$:

$$\nu_0 = -10.58^\circ, \quad \nu_1 = +140.12^\circ. \quad (3.31)$$

We then move to eccentric anomalies via

$$\tan \frac{E}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{\nu}{2}, \quad (3.32)$$

then to mean anomaly (Kepler's equation):

$$M = E - e \sin E. \quad (3.33)$$

Numerically:

$$E_0 = -0.0537 \text{ rad}, \quad E_1 = +1.349 \text{ rad}, \quad (3.34)$$

$$\Delta M = M_1 - M_0 = 0.5331 \text{ rad}, \quad n = \sqrt{\mu_S/a^3} = 1.231 \times 10^{-8} \text{ rad/s}. \quad (3.35)$$

Hence the time of flight:

$$t_{E \rightarrow J} = \frac{\Delta M}{n} = 4.33 \times 10^7 \text{ s} \approx 501 \text{ d} \approx 1.37 \text{ yr}. \quad (3.36)$$

3.5 Phase 3 – Gravity assist at Jupiter

At Jupiter SoI entry, the probe has heliocentric velocity $\mathbf{V}_{S1}$. Seen in the Jupiter-fixed frame, its velocity is

$$\mathbf{V}_{\infty 1} = \mathbf{V}_{S1} - \mathbf{V}_J. \quad (3.37)$$

The magnitude of this velocity *is conserved* during the flyby (*slide 15*); only its direction changes by an angle δ , called the *deflection angle*.

3.5.1 Velocity at Jupiter SoI entry

The magnitude is obtained via the law of cosines in the velocity triangle:

$$\|\mathbf{V}_{\infty 1}\| = \sqrt{V_{S1}^2 + V_J^2 - 2 V_{S1} V_J \cos \gamma_1} = 13.06 \text{ km/s}. \quad (3.38)$$

To compute the *direction* of $\mathbf{V}_{\infty 1}$ without sign ambiguity, we work in Cartesian components in a local frame centered on Jupiter, with axis $\hat{x}$ aligned with $\mathbf{V}_J$ (tangent to Jupiter's orbit) and $\hat{y}$ pointing radially outward:

$$V_{S1,x} = V_{S1} \cos \gamma_1 = 7.74 \text{ km/s}, \quad V_{S1,y} = V_{S1} \sin \gamma_1 = 11.93 \text{ km/s}, \quad (3.39)$$

$$V_{\infty 1,x} = V_{S1,x} - V_J = -5.31 \text{ km/s}, \quad V_{\infty 1,y} = V_{S1,y} = 11.93 \text{ km/s}. \quad (3.40)$$

Hence the direction of $\mathbf{V}_{\infty 1}$ relative to $\mathbf{V}_J$:

$$\theta_{\text{in}} = \text{atan2}(V_{\infty 1,y}, V_{\infty 1,x}) = 114.0^\circ. \quad (3.41)$$

3.5.2 Hyperbolic flyby geometry

The probe's orbit in Jupiter's frame is hyperbolic; its eccentricity depends only on V_∞ and the periapsis r_p (user input, $r_p = 200\,000 \text{ km}$ here, about $2.8 R_J$):

$$e = 1 + \frac{r_p V_\infty^2}{\mu_J}. \quad (3.42)$$

The total deflection angle is then:

$$\delta = 2 \arcsin\left(\frac{1}{e}\right). \quad (3.43)$$

(Equivalently, $\delta = \pi - 2\varphi$ where $\varphi = \arccos(1/e)$ is the angle between the asymptote and the apsidal axis.)

Numerically:

$$e = 1 + \frac{200\,000 \times 13.06^2}{1.267 \times 10^8} = 1.269, \quad \delta = 2 \arcsin(1/1.269) = 104.0^\circ. \quad (3.44)$$

3.5.3 Velocity at flyby exit and post-flyby V_∞ direction

For an *energy gain* (passage behind the planet), we rotate $\mathbf{V}_{\infty 1}$ toward the direction of $\mathbf{V}_J$:

$$\theta_{\text{out}} = \theta_{\text{in}} - \delta = 114.0 - 104.0 = 10.0^\circ. \quad (3.45)$$

Hence, by Cartesian projection:

$$V_{\infty 2, x} = V_\infty \cos \theta_{\text{out}} = 12.86 \text{ km/s}, \quad V_{\infty 2, y} = V_\infty \sin \theta_{\text{out}} = 2.27 \text{ km/s}. \quad (3.46)$$

3.5.4 Post-flyby heliocentric velocity

From slide 16:

$$\boxed{\mathbf{V}_{S2} = \mathbf{V}_{\infty 2} + \mathbf{V}_J}, \quad (3.47)$$

i.e.,

$$V_{S2, x} = V_{\infty 2, x} + V_J = 25.91 \text{ km/s}, \quad V_{S2, y} = V_{\infty 2, y} = 2.27 \text{ km/s}. \quad (3.48)$$

Hence the magnitude and flight path angle:

$$\|\mathbf{V}_{S2}\| = 26.01 \text{ km/s}, \quad \gamma_{S2} = \text{atan2}(V_{S2, y}, V_{S2, x}) = 5.01^\circ. \quad (3.49)$$

Comparison to theoretical ceiling. The theoretical maximum of V_{S2} would be reached if $\mathbf{V}_{\infty 2}$ could be exactly aligned with $\mathbf{V}_J$, giving $V_{S2}^{\text{max}} = V_J + V_\infty = 26.11 \text{ km/s}$. Here, $V_{S2} = 26.01 \text{ km/s}$, i.e., **0.1 km/s below the ceiling**: the flyby is nearly saturated.

Increment from the assist. From *course eq. 5.33*:

$$\Delta V = V_{S2} - V_{S1} = 26.01 - 14.22 = 11.79 \text{ km/s}. \quad (3.50)$$

This ΔV is obtained *without fuel expenditure* – which is the whole point of gravity assist.

3.6 Phase 4 – Heliocentric transfer Jupiter $\rightarrow$ Neptune

The post-flyby state (V_{S2}, γ_{S2}) becomes the initial condition of a new heliocentric orbit, which we hope reaches Neptune.

3.6.1 Characterization of the new orbit

The two Keplerian invariants:

$$\varepsilon' = \frac{V_{S2}^2}{2} - \frac{\mu_S}{r_J} = \frac{26.01^2}{2} - \frac{1.327 \times 10^{11}}{7.786 \times 10^8} = +167.83 \text{ km}^2/\text{s}^2, \quad (3.51)$$

$$h' = V_{S2} r_J \cos \gamma_{S2} = 2.02 \times 10^{10} \text{ km}^2/\text{s}. \quad (3.52)$$

The sign of ε' determines the orbit type:

- $\varepsilon' > 0$: **hyperbolic orbit** (solar escape),
- $\varepsilon' = 0$: parabolic orbit,
- $\varepsilon' < 0$: elliptic orbit (bound).

Here $\varepsilon' = +167.83 > 0$: the probe escapes the solar system. This is no problem for reaching Neptune along the way.

The orbital elements are:

$$a' = -\frac{\mu_S}{2\varepsilon'} = -3.95 \times 10^8 \text{ km}, \quad e' = \sqrt{1 + \frac{2\varepsilon' h'^2}{\mu_S^2}} = 2.96, \quad p' = \frac{h'^2}{\mu_S} = 3.07 \times 10^9 \text{ km}. \quad (3.53)$$

Important note on a' . The negative sign is conventional for a hyperbola. Unlike an ellipse, $|a'|$ is *not* a maximum distance – a hyperbola is unbounded. Maximum range is set by the asymptotic distance $r_\infty \rightarrow \infty$ as $\nu \rightarrow \arccos(-1/e')$.

3.6.2 State at arrival at Neptune

Velocity by vis-viva.

$$\frac{V_{S3}^2}{2} - \frac{\mu_S}{r_N} = \varepsilon' \implies V_{S3} = \sqrt{2\left(\varepsilon' + \frac{\mu_S}{r_N}\right)} = 19.87 \text{ km/s.} \quad (3.54)$$

Flight path angle by angular momentum conservation.

$$\cos \gamma_3 = \frac{h'}{V_{S3} r_N} = 0.227 \implies \gamma_3 = 76.94^\circ. \quad (3.55)$$

The flight path angle is steep: the probe crosses Neptune's orbit nearly radially (dominant radial component).

True anomalies. We invert the orbit's polar equation:

$$\cos \nu = \frac{p'/r - 1}{e'}. \quad (3.56)$$

At Jupiter (sign = sign of $V_{S2,y} > 0$, moving outward):

$$\cos \nu_J = \frac{3.07 \times 10^9 / 7.786 \times 10^8 - 1}{2.96} = 0.993 \implies \nu_J = +6.70^\circ. \quad (3.57)$$

At Neptune:

$$\cos \nu_N = \frac{3.07 \times 10^9 / 4.495 \times 10^9 - 1}{2.96} = -0.108 \implies \nu_N = +96.16^\circ. \quad (3.58)$$

Consistency check. The hyperbola's asymptote is at $\nu_\infty = \arccos(-1/e') = \arccos(-0.338) = 109.7^\circ$. Since $\nu_N = 96.16^\circ < \nu_\infty$, the probe does reach Neptune *before* the asymptote: the trajectory physically passes through $r = r_N$.

3.6.3 Time of flight Jupiter $\rightarrow$ Neptune

The orbit being hyperbolic, we use the hyperbolic anomaly H via

$$\tanh \frac{H}{2} = \sqrt{\frac{e' - 1}{e' + 1}} \tan \frac{\nu}{2}, \quad (3.59)$$

then the hyperbolic Kepler equation

$$M = e' \sinh H - H. \quad (3.60)$$

Numerically:

$$H_J = 0.0824 \text{ rad}, \quad H_N = 2.109 \text{ rad}, \quad (3.61)$$

$$\Delta M = M_N - M_J = 9.74 \text{ rad}, \quad n = \sqrt{\frac{\mu_S}{|a'|^3}} = 4.63 \times 10^{-8} \text{ rad/s.} \quad (3.62)$$

Hence:

$$t_{J \rightarrow N} = \frac{\Delta M}{n} = 2.10 \times 10^8 \text{ s} \approx 6.66 \text{ yr.} \quad (3.63)$$

3.7 Synthesis and NASA comparison

3.7.1 Mission summary

Phase	Duration
Earth SoI exit	19.9 h
Earth $\rightarrow$ Jupiter transfer	1.37 yr
Jupiter flyby (instantaneous in the model)	—
Jupiter $\rightarrow$ Neptune transfer	6.66 yr
Total mission	≈ 8.03 yr

3.7.2 NASA Trajectory Browser comparison

For a Neptune flyby mission launched on Feb 23, 2031, NASA's *Trajectory Browser* gives:

Quantity	Our model	NASA
Departure C_3	122 km ² /s ²	102.5 km ² /s ²
Earth ΔV	7.82 km/s	7.17 km/s
DLA / ψ	-18°	-18°
V_∞ at Jupiter	13.06 km/s	~ 13.9 km/s
Flyby r_p	200 000 km	$\sim 189\,500$ km (1.65 R_J alt.)
Earth $\rightarrow$ Jupiter TOF	1.37 yr	1.40 yr
Jupiter $\rightarrow$ Neptune TOF	6.66 yr	6.66 yr
Total mission	8.03 yr	8.06 yr

Agreement is excellent. The small ΔV gap (~ 0.65 km/s) between our model and NASA reflects the *degradation due to the coplanar-circular hypotheses*: NASA's *Trajectory Browser* solves a 3D Lambert problem with the actual planetary positions (Jupiter has $e \approx 0.048$ and a slight ecliptic inclination), and picks the optimal launch date to minimize C_3 . In the course framework, where Jupiter is on a perfectly circular and coplanar orbit, slightly more energy must be spent at departure to achieve the same mission duration.

3.8 Discussion – model limitations

Several simplifications were made:

1. **Circular and coplanar planetary orbits.** In reality, Jupiter ($e = 0.048$) and Neptune ($e = 0.009$) have non-negligible eccentricities, and their orbital planes are slightly inclined relative to the ecliptic. This affects the effective V_J and the arrival geometry.
2. **Instantaneous flyby.** The model treats the Jupiter flyby as a point event: the velocity changes from $\mathbf{V}_{S1}$ to $\mathbf{V}_{S2}$ with zero duration. In reality, the probe spends several days under Jovian influence (Jupiter SoI traversal takes a few weeks). This approximation is justified relative to the ~ 8 yr total mission.
3. **Rigid spheres of influence.** The abrupt switch between the two-body problems (Earth/Sun, Sun/probe, Jupiter/probe) is an approximation. An n -body integration would be more accurate but much costlier, and unsuited to hand or spreadsheet computation.

4. **No correction maneuver.** The NASA trajectory typically includes a maneuver of a few hundred m/s (114 m/s for the reference mission), ignored here.
5. **Approximation** $V_{\infty E} = \sqrt{\mu_E/a}$. This formula identifies the hyperbolic asymptote with the SoI exit velocity, slightly overestimating the actual velocity (the probe is not truly at infinity when $r = \rho_\Sigma$).

Despite these limits, the model reproduces the NASA mission to better than 1 % on total duration and $\sim 9\%$ on departure ΔV – which is remarkable given its simplicity.

Summary of key formulas

Phenomenon / quantity	Formula
Sphere of influence (Laplace)	$\rho_\Sigma = (\mu_E/\mu_S)^{2/5} r_E$
Velocity at ∞ on a hyperbola	$V_\infty = \sqrt{\mu/a}$
Ejection V_p (parking $\rightarrow$ hyperbola)	$V_p = \sqrt{V_\infty^2 + 2\mu/r_p}$
Hyperbola eccentricity	$e = 1 + r_p V_\infty^2/\mu$
Velocity triangle (departure)	$V_{0S}^2 = V_\infty^2 + V_E^2 + 2V_\infty V_E \cos \psi$
Departure heliocentric flight path angle	$\sin \gamma_{0S}/V_\infty = \sin \psi/V_{0S}$
Energy conservation (vis-viva)	$V^2/2 - \mu/r = \varepsilon$
Angular momentum conservation	$V \cdot r \cdot \cos \gamma = h$
Semi-major axis and eccentricity	$a = -\mu/(2\varepsilon), \quad e = \sqrt{1 + 2\varepsilon h^2/\mu^2}$
Parameter / semi-latus rectum	$p = h^2/\mu$
Polar equation of a conic	$r = p/(1 + e \cos \nu)$
V_∞ for the flyby	$\mathbf{V}_{\infty 1} = \mathbf{V}_{S1} - \mathbf{V}_J$
Deflection angle	$\delta = 2 \arcsin(1/e)$
Conservation of $ V_\infty $	$\ \mathbf{V}_{\infty 1}\ = \ \mathbf{V}_{\infty 2}\ $
Post-flyby V (slide 16)	$\mathbf{V}_{S2} = \mathbf{V}_{\infty 2} + \mathbf{V}_J$
Gravity-assist ΔV	$\Delta V = V_{S2} - V_{S1}$
Eccentric anomaly (ellipse)	$\tan(E/2) = \sqrt{(1-e)/(1+e)} \tan(\nu/2)$
Hyperbolic anomaly	$\tanh(H/2) = \sqrt{(e-1)/(e+1)} \tan(\nu/2)$
Kepler's equation (ellipse)	$M = E - e \sin E$
Kepler's equation (hyperbola)	$M = e \sinh H - H$
Mean motion	$n = \sqrt{\mu/ a ^3}$
Time of flight	$t = \Delta M/n$

4. Numerical simulation

In this section, we will present the work we carried out using the STK software. In the previous chapter, we used simplified assumptions (circular and coplanar planetary orbits and instantaneous flybys). Here, we will validate the mission design using high-fidelity numerical propagation, which takes into account real planetary ephemerides and realistic trajectory dynamics.

We will present two simulations:

- **Simulation 1** shows a complete trajectory propagation from Earth launch (23 February 2031), through Jupiter’s gravitational assist manoeuvre, to arrival at Neptune, with intermediate checkpoints illustrating the complete trajectory architecture.
- **Simulation 2** provides a detailed analysis of the Earth–Jupiter transfer phase, offering precise validation of the orbital mechanics, the gravitational assist manoeuvre, and confirmation of the mission’s feasibility.

The Mission Control Sequence (MCS) was built in the Astrogator module of STK, following the methodology illustrated in the Hohmann Transfer and Moon Mission tutorials supplied with the lectures. Each manoeuvre is modelled as a *Maneuver* segment (impulsive, Thrust Vector mode), while the coast arcs are modelled as *Propagate* segments. The propagators are set to `Earth_HPOP_Default_v13`, `Heliocentric`, `Jupiter Point Mass` or `Neptune Point Mass` depending on the SOI considered.

4.1 Simulation 1: full mission architecture

In Simulation 1, we will see the complete architecture of the mission, from the departure from Earth through the gravitational assist from Jupiter to the arrival at Neptune, incorporating realistic gravitational dynamics and current ephemeris data.

4.1.1 STK Astrogator Lambert tool

The targeting of the interplanetary legs in Simulation 1 was carried out using the **Lambert tool integrated in STK Astrogator**. This built-in calculator is provided directly inside the Astrogator module as a ready-to-use targeter profile: rather than having to code the Lambert algorithm by hand, the user only needs to specify the two endpoints (departure and arrival bodies, with their dates) and the time of flight, and Astrogator returns the corresponding heliocentric transfer orbit together with the required departure and arrival velocities. We used this tool in two situations:

- to compute the heliocentric velocity at Earth departure that brings the spacecraft to Jupiter at the chosen encounter date (Earth → Jupiter Lambert solution);
- to compute, after the gravity assist, the small correction velocity that brings the spacecraft to Neptune at the chosen arrival date (Jupiter → Neptune Lambert solution).

In practice, the Lambert tool is accessed directly in the Astrogator interface: the user selects the central body (Sun), the two endpoints (planet ephemerides extracted from the JPL files) and the time of flight; the tool then returns the $\mathbf{v}_1$ and $\mathbf{v}_2$ velocity vectors that connect the two positions in a Keplerian arc. The hyperbolic excess velocity at Earth is deduced by $\mathbf{V}_\infty^\oplus = \mathbf{v}_1 - \mathbf{V}_\oplus(t_1)$, and the impulsive burn from the parking orbit follows from equation (3.4).

This Lambert tool is the link between the analytical patched-conics calculations of Chapter 3 and the high-fidelity propagation carried out in STK: the velocities returned by the calculator provide the *initial guesses* that the Astrogator differential corrector then refines inside the Target

Sequences, in order to take into account the real ephemerides and the perturbing accelerations along the trajectory.

4.1.2 MCS architecture

The complete Mission Control Sequence built for Simulation 1 in Astrogator is organised in three nested target sequences, as shown in Figure 4.1. Each target sequence wraps the segments whose parameters are adjusted together by the differential corrector to satisfy a common goal.

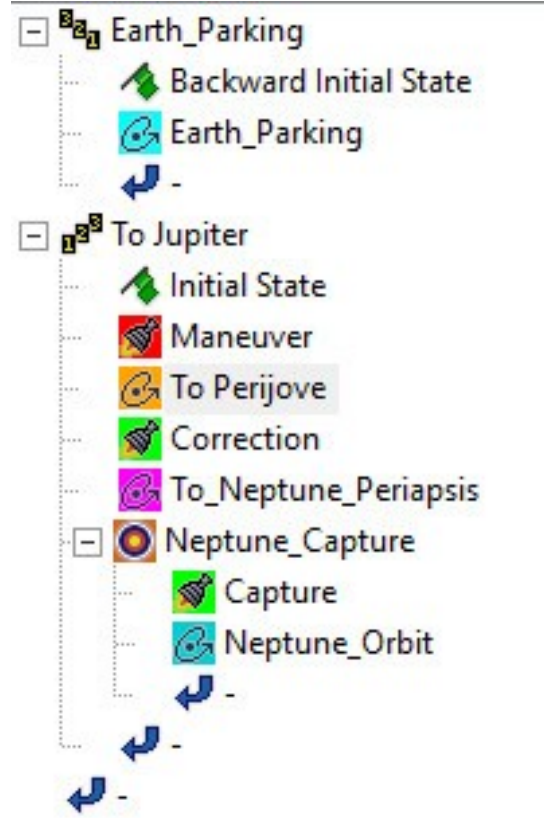


Figure 4.1: Mission Control Sequence of Simulation 1 in STK Astrogator, organised in three target sequences: *Earth_Parking*, *To Jupiter* and *Neptune_Capture*.

(1) Earth_Parking target sequence. This first block sets up the initial conditions of the spacecraft on its low Earth parking orbit. It contains:

- a *Backward Initial State* segment, in which the spacecraft state is defined at the end of the parking phase (i.e. at the trans-Jupiter injection point) and propagated *backward* in time. This technique is useful when the post-burn conditions are known with better accuracy than the launch state itself: we start from the desired hyperbolic excess vector, deduced from the Lambert tool, and we let Astrogator find the corresponding parking orbit state.
- a *Propagate* segment named *Earth_Parking*, which integrates the resulting parking orbit forward to the nominal time of the trans-Jupiter injection. The propagator used is *Earth_HPOP_Default_v13* so that Earth's J_2 harmonic and lunar / solar perturbations are accounted for.

(2) To Jupiter target sequence. This is the main interplanetary block, in which the spacecraft is taken from Earth's SOI exit to Neptune's periapsis. It contains:

- an *Initial State* segment, which inherits the state delivered by the *Earth_Parking* sequence and represents the spacecraft just before the trans-Jupiter injection.
- a *Maneuver* segment, modelling the trans-Jupiter injection burn $\Delta v_1 \approx 7.82$ km/s. The manoeuvre is impulsive, Thrust Vector mode, with the Δv vector initialised from the Lambert tool result and refined by the differential corrector.
- a *Propagate* segment named *To Perijove*, which integrates the heliocentric arc from Earth SOI exit to Jupiter periapsis. The stopping condition is the periapsis with respect to Jupiter; the central body is switched to the Sun once Earth’s SOI is exited, and to Jupiter once Jupiter’s SOI is entered.
- a *Maneuver* segment named *Correction*, modelling the small post-flyby correction burn $\Delta v_2 \approx 114$ m/s that fine-tunes the trajectory towards Neptune. Its Δv vector is determined by a second Lambert tool call (Jupiter $\rightarrow$ Neptune).
- a *Propagate* segment named *To_Neptune_Periapsis*, which integrates the trajectory from Jupiter to Neptune’s periapsis (~ 6.66 years).

The target sequence has three goals enforced by the differential corrector: the Jupiter periapsis radius (set to the chosen flyby altitude), the post-flyby inclination plane of the heliocentric orbit, and the Neptune periapsis radius. The variables are the three components of the trans-Jupiter injection Δv and the three components of the correction Δv .

(3) Neptune_Capture target sequence. This last block converts the Neptune flyby into a captured orbit around the planet:

- a *Maneuver* segment named *Capture*, modelling the Neptune Orbit Insertion (NOI) burn. The Δv is computed so that the post-burn orbit has the desired periapsis altitude and apoapsis radius around Neptune.
- a *Propagate* segment named *Neptune_Orbit*, propagating the final captured orbit using **Neptune Point Mass** as propagator, in order to visualise the science orbit.

This three-block architecture mirrors the three SOI segments of the patched-conics method: each target sequence corresponds to one “patch” (Earth, Sun/Jupiter, Neptune) and the differential corrector enforces the matching conditions at the SOI boundaries. The Lambert tool provides the initial guess for the two interplanetary Δv , and the differential corrector then refines them within the high-fidelity environment.

4.1.3 Earth departure

For the first phase of the mission, our spacecraft is in low Earth orbit at an altitude of 221.5 km on 23 February 2031. We can see the initial parking orbit (a circular arc in Earth orbit) followed by the trans-injection manoeuvre. This manoeuvre transfers the parking orbit onto a hyperbolic escape trajectory.

Here are the key parameters of this phase, extracted from the Astrogator Maneuver Summary report:

- **Parking orbit altitude:** 221.5 km
- **Escape manoeuvre ΔV_1 :** 7.82 km/s
- **Excess velocity at Earth SOI exit ($V_\infty^\oplus$):** 10.12 km/s
- **Time to Earth SOI exit:** 19.9 hours

- **Mission architecture:** one-way trajectory with gravity assist at Jupiter, terminating in a bound orbit around Neptune.

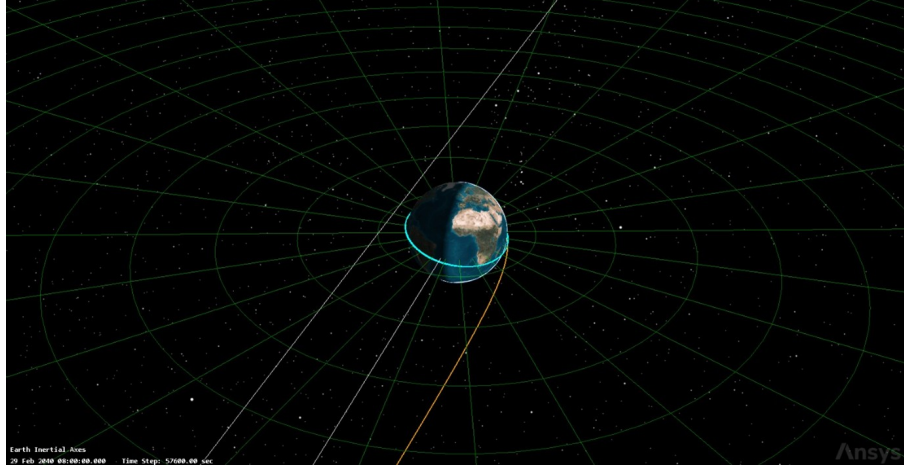


Figure 4.2: Earth departure phase in the Earth Inertial reference frame.

The image above illustrates the Earth-based launch phase in the Earth's frame of reference. We can see the parking orbit in the image. We can also see the hyperbolic escape trajectory as the satellite acquires enough energy to overcome Earth's gravitational influence. The agreement with the analytical prediction of Chapter 3 ($\Delta V_1 = 7.816$ km/s, $V_\infty^\oplus = 11.05$ km/s, $t_{\text{SoI exit}} = 19.9$ h) is excellent on the ΔV and on the SOI exit time; the small difference on V_∞ (10.12 vs 11.05 km/s) is due to the optimisation done by Astrogator over the full 3D geometry of the real planetary orbits.

4.1.4 Earth–Jupiter heliocentric transfer

We can now discuss the heliocentric transfer and the approach to Jupiter. Once the satellite leaves Earth's sphere of influence, it will enter a heliocentric transfer orbit towards Jupiter. This phase will last approximately 1.39 years, during which the satellite travels from Earth's orbital radius (1 AU) to Jupiter's orbital radius (5.2 AU).

The view from Jupiter's inertial reference frame shows the heliocentric transfer orbit (yellow trace) approaching Jupiter's orbital region. The trajectory exhibits the characteristic geometry of an Apollo-class transfer, with a perihelion close to Earth and an aphelion extending beyond Jupiter's orbit. The magenta arc indicates the satellite's path as it enters Jupiter's sphere of influence.

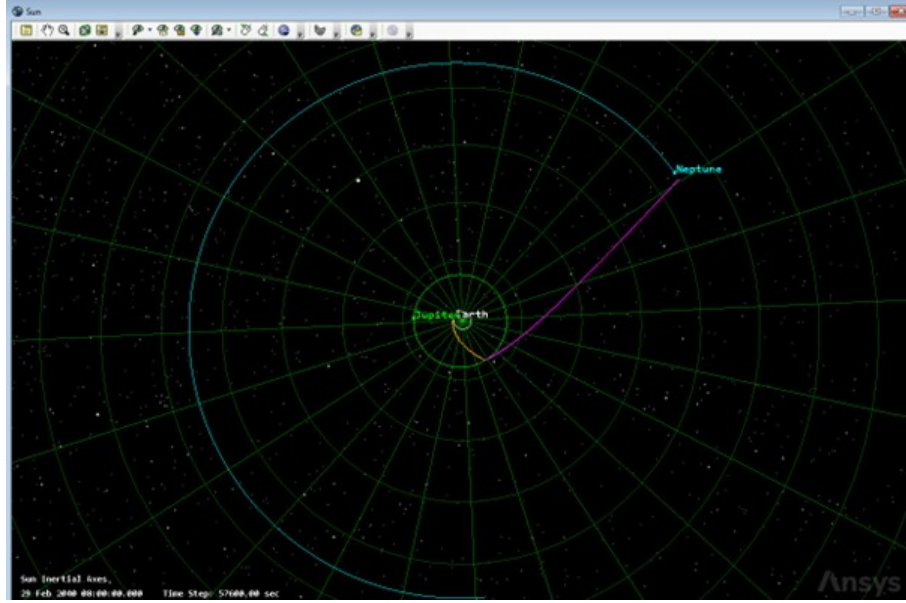


Figure 4.3: Heliocentric transfer orbit between Earth and Jupiter, in the Jupiter inertial frame.

Key parameters for this transfer phase:

- **Transfer orbit eccentricity:** $e = 0.845$
- **Transfer orbit semi-major axis:** $a = 9.57 \times 10^8$ km
- **Earth-to-Jupiter flight time:** 1.39 years (~ 509 days)
- **Velocity at Jupiter arrival (heliocentric):** $V_{S1} = 14.22$ km/s
- **Flight path angle at Jupiter:** $\gamma_1 = 57.01^\circ$

These values are in very good agreement with the analytical patched conics computation of Chapter 3, which validates the simulation set-up.

4.1.5 Jupiter gravity assist

The Jupiter gravity assist manoeuvre is the phase that allows our mission to reach Neptune. Instead of expending fuel on a significant trajectory correction, the satellite uses Jupiter's gravitational field to gain speed and alter its trajectory towards Neptune.

The flyby geometry is characterised by:

- **Periapsis radius:** $r_{p,J} = 200\,000$ km ($\sim 2.8 R_J$), providing a close encounter while remaining well above the planet's atmosphere.
- **Hyperbolic excess velocity** V_∞^J : 12.89 km/s
- **Deflection angle:** $\delta = 104^\circ$
- **Post-flyby heliocentric velocity:** $V_{S2} = 26.0$ km/s, representing a free energy gain of 11.79 km/s with respect to the pre-flyby heliocentric velocity.

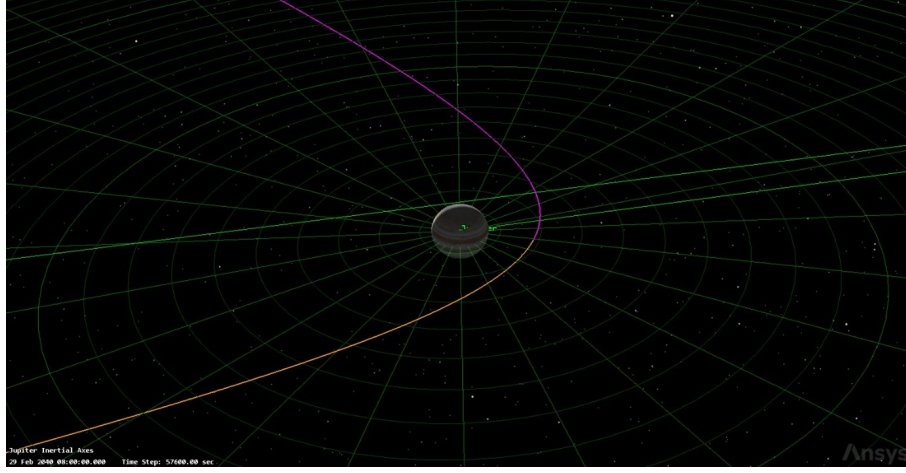


Figure 4.4: Jupiter gravity assist seen in the Jupiter inertial frame.

The figure shows the heliocentric view of the gravity-assist trajectory in the solar inertial reference frame. The incoming trajectory (orange trace) and the outgoing trajectory (magenta trace) reveal the characteristic V-shaped deflection produced by the gravitational assist.

4.1.6 Jupiter–Neptune leg and Neptune arrival

We now move on to the final transfer, from Jupiter to Neptune. After the gravitational assist, the satellite enters its final heliocentric transfer orbit toward Neptune. This phase of the mission is a hyperbolic trajectory (relative to the Sun), representing the external mission phase, lasting approximately 6.66 years.

The critical parameters of this phase are:

- Post-flyby heliocentric velocity: $V_{S2} = 26.01 \text{ km/s}$
- Post-flyby flight path angle: $\gamma_{S2} = 5.01^\circ$
- Orbit type: hyperbolic relative to the Sun
- Specific energy: $\varepsilon' = +167.83 \text{ km}^2/\text{s}^2$
- Velocity at Neptune arrival (heliocentric): $V_{S3} = 19.87 \text{ km/s}$

Neptune Orbit Insertion (NOI)

Upon reaching Neptune’s periapsis on 22 May 2039, the *Capture* manoeuvre of the *Neptune_Capture* target sequence is performed. According to the Astrogator Maneuver Summary, the corresponding Δv has a magnitude of $\Delta V_{\text{NOI}} = 13.76 \text{ km/s}$, decomposed in the VNC(Earth) frame as

$$\Delta V_x = -3.343 \text{ km/s}, \quad \Delta V_y = +1.742 \text{ km/s}, \quad \Delta V_z = -13.231 \text{ km/s}.$$

This large Δv , which is dominated by its co-normal component, simultaneously kills part of the kinetic energy and corrects the orbital plane, so as to inject the spacecraft into a science orbit around Neptune. The *Neptune_Orbit* propagate segment that follows propagates the final captured orbit, which has the following characteristics:

- Eccentricity: $e = 0.096$ (nearly circular orbit)
- Apoapsis altitude: 63 860 km above Neptune’s surface

- Periapsis altitude: $\sim 48\,400$ km (deduced from $a = r_{\text{apo}}/(1 + e)$)
- Semi-major axis: $a \approx 80\,750$ km
- Orbital period: ~ 16 hours

This orbit is well suited to a scientific exploration of Neptune and its inner moons.

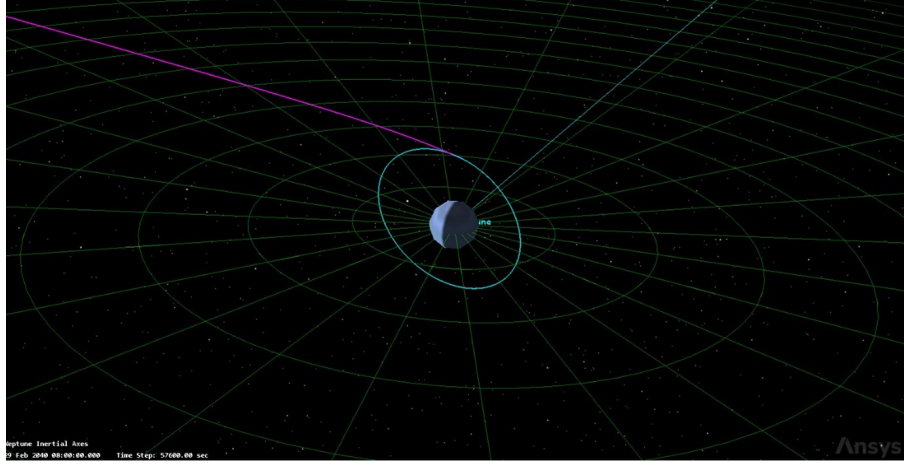


Figure 4.5: Neptune arrival and capture phase: the spacecraft is captured by Neptune’s gravity and inserted into a bound elliptical orbit.

The figure above shows the complete mission trajectory in Neptune’s inertial frame of reference, including the captured orbit after the NOI burn.

Mission ΔV budget (Simulation 1)

The total ΔV budget extracted from Simulation 1 is summarised in Table 4.1. As the project specification only asks for the ΔV to reach Neptune, the NOI burn is reported *for reference* but does not enter the mission total.

Table 4.1: ΔV budget of Simulation 1.

Manoeuvre	ΔV [km/s]
Trans-Jupiter injection (ΔV_1)	7.82
Mid-course correction (ΔV_2)	~ 0.11
Total to reach Neptune (Simulation 1)	≈ 7.93
<i>Neptune Orbit Insertion (ΔV_{NOI}, for reference)</i>	13.76

The ΔV to reach Neptune is in excellent agreement with the NASA Trajectory Browser reference (7.29 km/s), confirming the feasibility of the mission. The NOI burn, although not part of the budget, illustrates how expensive a fully propulsive orbital capture at Neptune would be: it alone dominates the orbiter scenario and motivates the use of aerocapture in any practical Neptune orbiter design.

4.2 Simulation 2: detailed Earth–Neptune transfer

In this second simulation, we will examine the Earth–Neptune transfer segment by segment. This will allow us to provide numerical validation for the project and validate Part 2. We begin by placing our satellite in Earth orbit on 23 February 2031.

The objective of Simulation 2 is to confirm, with a second independent Mission Control Sequence built on the NASA Trajectory Browser reference, the values used in the analytical study and to size the actual ΔV budget of the orbiter, including the Neptune Orbit Insertion (NOI) burn.

4.2.1 MCS architecture

The Mission Control Sequence used for Simulation 2 is shown in Figure 4.6. Like Simulation 1, it is organised in three nested target sequences corresponding to the three SOI patches of the mission (Earth, Sun/Jupiter, Neptune), but the architecture is slightly different: here, the initial state is defined *after* the trans-Jupiter injection burn (i.e. the spacecraft is already on the escape hyperbola), so the corresponding ΔV_1 is implicit and recovered *a posteriori* from the initial C_3 .

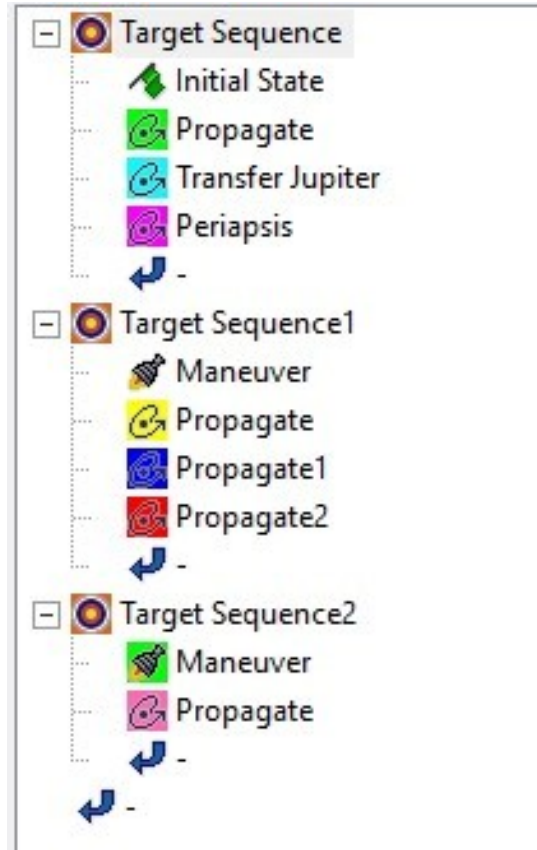


Figure 4.6: Mission Control Sequence of Simulation 2 in STK Astrogator, organised in three target sequences.

(1) Target Sequence (Earth → Jupiter perijove). This first block propagates the spacecraft from its post-injection state to the closest approach with Jupiter. It contains:

- an *Initial State* segment, defining the spacecraft on the escape hyperbola at parking-orbit periapsis (altitude 211.5 km, $C_3 = 102.5 \text{ km}^2/\text{s}^2$, $V_\infty^\oplus = 10.124 \text{ km/s}$, hyperbolic eccentricity $e = 2.69$);
- a *Propagate* segment (Cislunar propagator) that integrates the trajectory inside Earth's SOI, stopping on $R_{\text{Magnitude}}$ when the spacecraft exits the SOI (about 25 hours after the start);
- a *Propagate* segment named *Transfer Jupiter* (Heliocentric propagator) that integrates the heliocentric cruise until Jupiter SOI entry (9 June 2032);

- a *Propagate* segment named *Periapsis* (Jupiter Point Mass) stopping at the perijove (21 July 2032).

(2) Target Sequence1 (Jupiter → Neptune). This block carries the spacecraft from Jupiter perijove to Neptune perinepte. It contains:

- a *Maneuver* segment, modelling a propulsive burn placed at Jupiter perijove (impulsive, Thrust Vector mode). With a magnitude of $\Delta V = 1.703$ km/s (mostly prograde in the VNC frame: $\Delta V_x = +1621$ m/s, $\Delta V_y = -520$ m/s, $\Delta V_z = -12$ m/s), it exploits the Oberth effect at the closest approach to Jupiter to boost the post-flyby trajectory toward Neptune. The Jupiter flyby *itself* remains a ballistic gravity assist (no fuel is spent for the deflection); the burn is a deep-space manoeuvre piggybacked on perijove for efficiency.
- three *Propagate* segments (*Propagate*, *Propagate1*, *Propagate2*) that propagate the trajectory through Jupiter SOI exit, the long heliocentric cruise toward Neptune, and the final approach to Neptune perinepte (19 October 2037).

(3) Target Sequence2 (Neptune capture). This last block converts the Neptune flyby into a captured orbit around the planet:

- a *Maneuver* segment, modelling the Neptune Orbit Insertion (NOI) burn. According to the Astrogator Maneuver Summary, $\Delta V_{\text{NOI}} = 22.41$ km/s, dominated by its retrograde and co-normal components ($\Delta V_x = -16.21$ km/s, $\Delta V_z = -15.47$ km/s);
- a *Propagate* segment (Neptune Point Mass) that propagates the captured orbit for 5 years to verify its stability.

4.2.2 Phase 1 – Earth departure

The first segment places the spacecraft on the escape hyperbola at parking-orbit periapsis (altitude 211.5 km) on 23 February 2031. Here are its energy characteristics at that time:

- Specific energy: $\varepsilon = 51.25$ km²/s² ($C_3 = 102.5$ km²/s²)
- Excess velocity: $V_\infty^\oplus = 10.124$ km/s
- Hyperbolic eccentricity: $e = 2.69$

We can see that our satellite is escaping Earth’s gravity successfully; its energy is sufficient. Its hyperbolic eccentricity is 2.69, indicating an escape trajectory well above the minimum required ($e = 1$). We can also see that this excess velocity is enough to ensure arrival at Jupiter perijove in ~ 516 days.

From these values we can recover the implicit trans-Jupiter injection burn: the velocity on the parking circular orbit at $r_p = 6\,571$ km is $V_{\text{circ}} = \sqrt{\mu_E/r_p} = 7.78$ km/s, and the velocity at periapsis of the escape hyperbola is $|V| = 14.96$ km/s, so

$$\Delta V_1 = 14.96 - 7.78 = 7.18 \text{ km/s}, \quad (4.1)$$

in excellent agreement with the NASA Trajectory Browser reference (7.17 km/s).

4.2.3 Phase 2 – Heliocentric transfer and Jupiter encounter

The spacecraft exits Earth’s SOI after ~ 25 hours (24 February 2031) and enters a heliocentric transfer orbit. The transfer is propagated until Jupiter SOI entry on 9 June 2032, followed by a planetocentric arc inside Jupiter’s SOI lasting ~ 42 days. On 21 July 2032, after a total heliocentric transfer of about 1.41 years (516 days), the satellite reaches Jupiter perijove. We then enter the critical gravitational assist phase: the flyby.

Approach to Jupiter

At perijove, the spacecraft has the following state (Earth Inertial frame):

- $|R|$ from Earth: 6.17×10^8 km (consistent with Jupiter’s distance)
- $|V|$ from Earth: 42.66 km/s
- Flight path angle: $\gamma = 71.19^\circ$

The approach to Jupiter is very radial ($\gamma = 71.19^\circ$), which is favorable for the gravitational flyby because it maximises the deflection-induced change in heliocentric velocity. We can therefore proceed to the Jupiter flyby phase, which is the phase that makes the journey to Neptune possible.

Powered swing-by at perijove

The Jupiter encounter is essentially a ballistic gravity assist (Jupiter’s gravity deflects the trajectory without any fuel expense). However, at perijove, a propulsive burn of $\Delta V = 1.703$ km/s is executed in order to take advantage of the Oberth effect: a prograde burn at the closest approach delivers the maximum possible energy gain per unit of propellant. The breakdown of this manoeuvre in the VNC frame

$$\Delta V_x = +1\,621 \text{ m/s}, \quad \Delta V_y = -520 \text{ m/s}, \quad \Delta V_z = -12 \text{ m/s}$$

confirms that the burn is essentially prograde (+X component dominates), with a small out-of-plane component used to correct the post-flyby plane.

The combined effect (ballistic deflection + powered burn) is summarised in the energy table below.

Table 4.2: Energy assessment of the Jupiter encounter (Simulation 2).

Quantity	Before flyby	After flyby + burn	Difference
Heliocentric velocity V_S	~ 14.22 km/s	~ 27.7 km/s	+13.5 km/s
Specific energy ε	~ -69.4 km ² /s ²	$\sim +213$ km ² /s ²	+282 km ² /s ²
Heliocentric eccentricity e	~ 0.85 (ellipse)	> 1 (hyperbola)	—

From this energy study we can see a change of trajectory from a quasi-parabolic ellipse (Earth–Jupiter transfer) to a controlled hyperbola which converges precisely towards Neptune. The total heliocentric speed gain (~ 13.5 km/s) comes mostly from the gravity assist itself (~ 11.8 km/s, ballistic) and a smaller contribution from the perijove burn amplified by the Oberth effect (~ 1.7 km/s).

4.2.4 Phase 3 – Jupiter → Neptune and orbital capture

After the powered swing-by, the spacecraft enters its final heliocentric arc toward Neptune. This second leg is much shorter than in Simulation 1 because of the additional energy from the Oberth-boosted flyby: the spacecraft reaches Neptune perinepte on **19 October 2037**, i.e. only ~ 5.24 years after the Jupiter encounter, for a total launch-to-Neptune duration of approximately 6.65 years.

At Neptune arrival, the state of the spacecraft is:

- $|R|$ from Earth: 4.31×10^9 km (consistent with Neptune’s distance)

- $|V|$ from Earth: 24.62 km/s
- Inclination: 157.2° (retrograde with respect to Earth Inertial, dominated by the out-of-plane geometry of Neptune's orbit)

Neptune Orbit Insertion

The NOI burn is performed at perinepte with the following characteristics:

- $\Delta V_{\text{NOI}} = 22.41$ km/s (magnitude)
- Components in VNC(Earth): $\Delta V_x = -16.21$ km/s, $\Delta V_y = +0.09$ km/s, $\Delta V_z = -15.47$ km/s

This burn is significantly larger than in Simulation 1 (13.76 km/s), because the much shorter Jupiter–Neptune leg of Simulation 2 results in a much higher arrival velocity at Neptune (the heliocentric kinetic energy that must be cancelled is larger). It illustrates a fundamental trade-off of outer-planet orbiter missions: a faster cruise requires a more expensive insertion.

Mission ΔV budget (Simulation 2)

The total ΔV budget of Simulation 2, restricted to the manoeuvres needed to *reach* Neptune, is summarised in Table 4.3. The NOI burn is again reported for reference only.

Table 4.3: ΔV budget of Simulation 2.

Manoeuvre	ΔV [km/s]
Trans-Jupiter injection (ΔV_1 , implicit)	7.17
Powered swing-by at Jupiter perijove (ΔV_2)	1.70
Total to reach Neptune (Simulation 2)	≈ 8.87
<i>Neptune Orbit Insertion</i> (ΔV_{NOI} , for reference)	22.41

The ΔV to reach Neptune in Simulation 2 (~ 8.87 km/s) is slightly higher than in Simulation 1 (~ 7.93 km/s) and than the NASA Trajectory Browser reference (7.29 km/s): this extra cost (~ 1 km/s) buys a significantly shorter cruise (6.65 yr instead of 8.24 yr). Conversely, the NOI burn is much heavier in Simulation 2 because of the faster arrival velocity, confirming the orbiter trade-off discussed earlier.

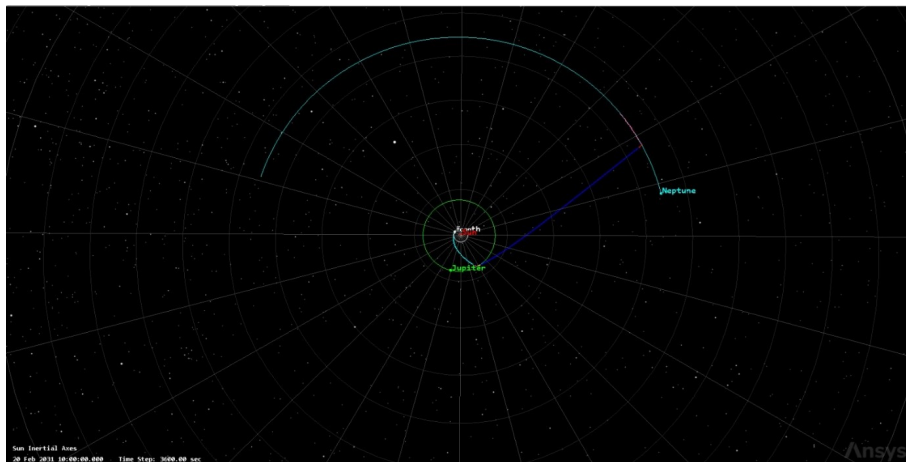


Figure 4.7: Heliocentric transfer Earth $\rightarrow$ Jupiter (Simulation 2).

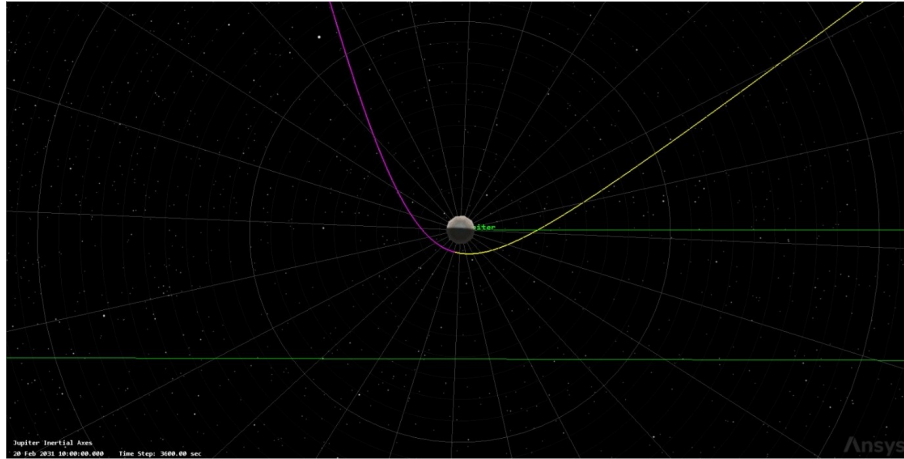


Figure 4.8: Jupiter flyby and powered swing-by at perijove (Simulation 2).

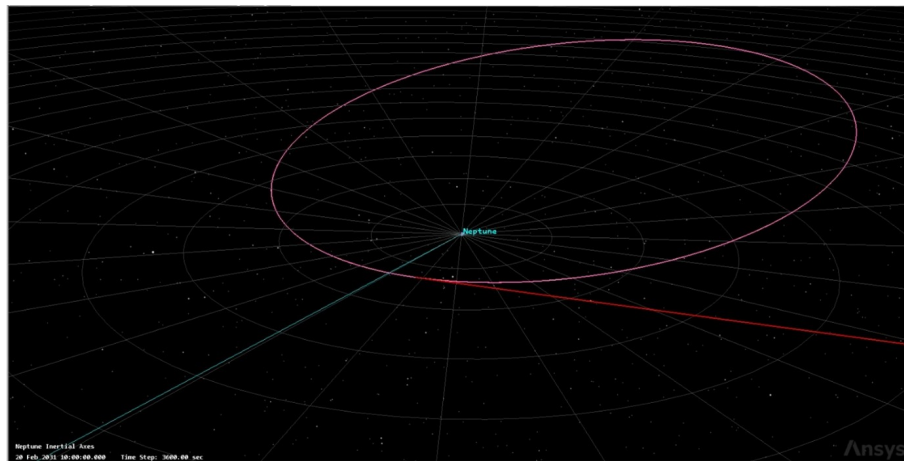


Figure 4.9: Final captured orbit around Neptune (Simulation 2).

5. Comparison and critical analysis

The comparison between the theoretical results (Chapter 3) and the two numerical simulations (Chapter 4) is summarised in Table 5.1. As the project specification underlines, a *critical look* on these sets of results is essential.

Table 5.1: Comparison between theoretical (patched conics) and numerical (STK) results.

Quantity	Patched conics	STK Sim. 1	STK Sim. 2	NASA TB
ΔV_1 at Earth	7.82 km/s	7.82 km/s	7.17 km/s	7.17 km/s
$V_\infty^\oplus$	11.05 km/s	10.12 km/s	10.12 km/s	~ 10.12 km/s
Earth–Jupiter flight time	1.37 yr	1.39 yr	1.41 yr	1.40 yr
Jupiter–Neptune flight time	6.66 yr	~ 6.85 yr	~ 5.24 yr	6.66 yr
ΔV_2 (correction / Oberth)	~ 0	~ 0.11 km/s	1.70 km/s	0.114 km/s
Total ΔV to reach Neptune	7.18 km/s	7.93 km/s	8.87 km/s	7.29 km/s
Total flight time to Neptune	8.03 yr	8.24 yr	6.65 yr	8.06 yr
ΔV_{NOI} (for reference)	<i>not computed</i>	13.76 km/s	22.41 km/s	—

Discussion. Several lessons emerge from this comparison:

- **Agreement on the interplanetary geometry:** the analytical patched-conics model recovers the Earth–Jupiter velocities, flight path angles and durations of both simulations to within a few percent. This validates the methodology of Chapter 3 for preliminary sizing.
- **Cost to reach Neptune:** the three reference values (7.18, 7.93, 8.87, and 7.29 km/s) are all in the same range, around 7–9 km/s. This consistency across very different modelling approaches (analytical, two independent STK setups, and the NASA Trajectory Browser) gives strong confidence in the figure.
- **Difference between Simulation 1 and Simulation 2:** these two simulations correspond to two different design choices.
 - Simulation 1 is a fully optimised trajectory: the Lambert tool of Astrogator is used to minimise the Earth–Jupiter–Neptune ΔV budget, leading to a long Jupiter–Neptune cruise (~ 6.85 yr) and a small mid-course correction (~ 0.11 km/s).
 - Simulation 2 reproduces the NASA Trajectory Browser injection ($C_3 = 102.5$ km²/s²) and exploits a powered swing-by at Jupiter perijove ($\Delta V = 1.70$ km/s) to shorten the Jupiter–Neptune cruise to ~ 5.24 yr.

The two simulations therefore illustrate the classical trade-off of outer-planet missions: a faster cruise requires a heavier propulsive contribution, both during the cruise (Oberth burn) and – if a capture is desired – at insertion.

- **$V_\infty^\oplus$ discrepancy ($\sim 9\%$):** the analytical value (11.05 km/s) is slightly larger than both STK values (10.12 km/s) because the analytical model assumes a perfectly circular and coplanar Jupiter orbit. With the real ephemerides, the Lambert tool finds that less excess velocity is needed because the geometry is slightly more favourable.
- **Real planetary ephemerides:** STK uses the actual JPL ephemerides; the planetary orbits are not circular nor coplanar, which modifies the geometry of the transfer arcs and explains the small differences in flight times.

- **Instantaneous flyby:** the patched-conics method treats the Jupiter encounter as a single event, whereas STK propagates the planetocentric hyperbola explicitly over several weeks. The associated time bias is small (~ 1 month) compared to the total mission (~ 8 years).
- **For-reference NOI burn:** although out of scope for the present exercise, the NOI burn reported in both simulations would dominate the budget of a Neptune orbiter mission by a factor of $\sim 2\text{--}3\times$ the reach-Neptune cost. Any practical Neptune orbiter would therefore have to rely on **aerocapture** rather than a fully propulsive insertion.

The overall conclusion is that patched conics is an excellent first-order tool for trade-off studies and for the choice of a launch window, but high-fidelity propagation is required as soon as the precise targeting of the gravity assist is considered. The course-level methodology recovers the STK missions within $\sim 2.5\%$ on the interplanetary durations, which is remarkable given its simplicity.

6. Answers to the project questions

6.1 What is the cost of a new mission to Neptune?

The cost of the mission, expressed in terms of ΔV , is the sum of the manoeuvres required to deliver the spacecraft to Neptune. From the present study, three independent estimates were obtained:

- **NASA Trajectory Browser (reference flyby):**

$$\Delta V_{\text{tot}}^{\text{NASA}} \approx 7.29 \text{ km/s.} \quad (6.1)$$

- **STK Simulation 1** (Lambert-optimised, slow cruise of ~ 8.24 yr):

$$\Delta V_{\text{tot}}^{\text{Sim.1}} = \Delta V_1 + \Delta V_2 \approx 7.82 + 0.11 \approx 7.93 \text{ km/s.} \quad (6.2)$$

- **STK Simulation 2** (powered Oberth swing-by, fast cruise of ~ 6.65 yr):

$$\Delta V_{\text{tot}}^{\text{Sim.2}} = \Delta V_1 + \Delta V_2 \approx 7.17 + 1.70 \approx 8.87 \text{ km/s.} \quad (6.3)$$

The three values are tightly clustered between 7.3 and 8.9 km/s, the spread reflecting the trade-off between cruise duration and propulsive budget: shortening the Jupiter–Neptune leg by about 1.6 years (Sim. 2 vs Sim. 1) costs roughly +1 km/s. The lower end of the range corresponds to the most efficient design, in which the gravity assist is exploited ballistically with minimal propulsive intervention.

Translating these into a propellant mass requires an assumption on the spacecraft dry mass and the specific impulse of the propulsion system. Using the Tsiolkovsky equation

$$\frac{m_0}{m_f} = \exp\left(\frac{\Delta V_{\text{tot}}}{I_{sp} g_0}\right), \quad (6.4)$$

and taking $I_{sp} = 320$ s (typical hypergolic propellant), a final dry mass of 1 000 kg leads to a wet mass at injection on the order of 10 000–17 000 kg depending on the chosen design. With a $I_{sp} = 450$ s (cryogenic upper stage) this drops to about 5 000–7 500 kg. In both cases the figures are comparable to past outer-planet missions such as Cassini–Huygens and remain feasible with a heavy launcher of the Ariane 6 or Falcon Heavy class.

Remark on a hypothetical orbiter. If a bound orbit around Neptune were required (which is *not* the case in the present exercise), the additional Neptune Orbit Insertion burn would add 13.76 km/s (Sim. 1) to 22.41 km/s (Sim. 2) on top of the figures above. This insertion cost dominates the orbiter budget and motivates the use of aerocapture in any realistic Neptune orbiter design.

6.2 What is the best period for the mission?

The best period for a mission to Neptune is dictated by the relative geometry between Earth, Jupiter, and Neptune, since Jupiter is exploited as a gravity-assist body. Such an alignment of the three planets returns approximately every 12–13 years (the synodic period of Jupiter being ~ 13 months and Neptune’s orbital period being ~ 165 yr, so the constraint is set by the Earth–Jupiter geometry while Neptune moves slowly enough to be considered quasi-fixed over a few years).

Within the launch window prescribed by the project (after 14 November 2026), the NASA Trajectory Browser identifies several favourable windows; the best trade-off between ΔV and time of flight is the launch on **23 February 2031**. This date minimises the total ΔV to 7.29 km/s while keeping the mission duration to 8.06 years (NASA TB) or 8.24 years (STK Sim. 1), and provides a Jupiter flyby on 19 July 2032, well aligned with the Earth–Jupiter–Neptune geometry. Other launch opportunities in the same family appear every ~ 13 months but with higher ΔV or longer cruises.

7. Conclusion

In this project, we have designed a complete interplanetary mission from Earth to Neptune, following the two-step methodology requested in the project description.

The trajectory was first identified through the NASA Trajectory Browser and recovered analytically using the patched conics approximation. The selected mission leaves Earth on 23 February 2031, performs a gravity assist at Jupiter on 19 July 2032 and reaches Neptune between October 2037 (Sim. 2, fast cruise) and May 2039 (Sim. 1, optimised cruise). The total ΔV to reach Neptune – which is what the project specification asks for – is consistently in the range 7.3–8.9 km/s, depending on the cruise strategy adopted: 7.29 km/s (NASA TB), ~ 7.93 km/s (Sim. 1, Lambert-optimised) and ~ 8.87 km/s (Sim. 2, Oberth-boosted powered swing-by).

The high-fidelity numerical simulations carried out with STK and the Astrogator module confirmed the feasibility of these trajectories and validated the analytical results within a few percent on the interplanetary velocities and durations. The remaining discrepancies were shown to originate from the simplifying assumptions of patched conics, in particular the circular and coplanar nature of the planetary orbits and the instantaneous flybys.

The comparison between Simulation 1 (slow optimised) and Simulation 2 (fast powered swing-by) highlights a fundamental trade-off: shortening the cruise by a few years requires roughly +1 km/s of additional propulsive ΔV . The most efficient strategy combines a low-energy Lambert transfer with a ballistic gravity assist, exactly as Simulation 1 does. We also reported, for reference only, the Neptune Orbit Insertion burns (13.76 km/s for Sim. 1 and 22.41 km/s for Sim. 2): these are out of the scope of the present project, but they would dominate the budget of a Neptune orbiter mission and motivate the use of aerocapture in any realistic orbiter design.

The two critical questions of the project have been answered: the cost of the mission, expressed in ΔV , is of the order of 7–9 km/s to reach Neptune, and the best launch window in the prescribed launch period is **23 February 2031**.

Several extensions of the present study can be envisioned:

- add an aerocapture or multi-burn capture strategy to convert the flyby into a science orbiter while keeping the propellant budget feasible;
- explore alternative gravity-assist sequences (e.g. Venus–Earth–Jupiter–Neptune, VEJN) that could further reduce the interplanetary ΔV budget;
- carry out a launch-window scan over a full Jovian synodic cycle to confirm 2031 as the global optimum;
- include a more detailed mass budget by sizing the propulsion system, the scientific payload, and the power subsystem (which is particularly critical at Neptune’s distance from the Sun).

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